🔑 **Answer Key — Teacher Copy · Lesson 7-8**

MSTAR PRACTICE WORKSHEET

Lesson 7-8: Ordered Pairs in All Four Quadrants

Grade 6 Mathematics · Standard 6.NOS.7

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

In which quadrant is the point (-4, 5)?

A. Quadrant I

B. Quadrant III

C. Quadrant II

D. Quadrant IV

The x-coordinate is negative (left) and the y-coordinate is positive (up). Left and up is Quadrant II.

Question 2 SELECTED RESPONSE — CORRECT: C

A point has a positive x-coordinate and a negative y-coordinate. Which quadrant is it in?

A. Quadrant I

B. Quadrant II

C. Quadrant IV

D. Quadrant III

Positive x (right) and negative y (down) = Quadrant IV (bottom-right).

Question 3 SELECTED RESPONSE — CORRECT: C

Which point is located on the x-axis?

A (-4, -4)

B (0, 4)

C (4, 0)

D (4, 4)

A point is on the x-axis when its y-coordinate is 0. (4, 0) is on the x-axis.

Question 4 SELECTED RESPONSE — CORRECT: D

Point A is at (-5, 3). If you move 8 units to the right, what are the new coordinates?

A (-13, 3)

B (-5, 11)

C (-5, -5)

D (3, 3)

Moving right means adding to the x-coordinate: -5 + 8 = 3. The y stays the same. New point: (3, 3).

Question 5 SELECTED RESPONSE — CORRECT: D

In which quadrant is the point (-3, 5)?

A Quadrant I

B Quadrant III

C Quadrant IV

D Quadrant II

Negative x with positive y places the point in Quadrant II.

Question 6 SELECTED RESPONSE — CORRECT: D

What are the coordinates of a point 4 units right and 2 units down from the origin?

A (-4, 2)

B (-2, 4)

C (2, -4)

D (4, -2)

Right is +x and down is -y, so the point is (4, -2).

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

In which quadrant does the point (–4, 3) lie?

A Quadrant II

B Quadrant I

C Quadrant III

D Quadrant IV

A negative x with a positive y is the upper-left region, which is Quadrant II.

- **B:** Quadrant I holds two positive coordinates. Here x is negative.
- **C:** Quadrant III holds two negative coordinates. Here y is positive.
- **D:** Quadrant IV has x positive and y negative. This point is the other way round.

Part B — correct answer: C

What sign pattern identifies Quadrant IV?

A (−, −)

B (−, +)

C (+, −)

D (+, +)

Quadrant IV is the lower-right region: x is positive (right of the y-axis) and y is negative (below the x-axis).

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about the four quadrants:

- **A** Quadrant I contains points with both coordinates positive.
- **B** Quadrant III contains points with both coordinates negative.
- **C** A point on an axis belongs to no quadrant.
- **D** The quadrants are numbered clockwise starting from the upper right.
- **E** Quadrant II contains points of the form (−, +).

A, B, C, and E are true. D is false — the quadrants are numbered COUNTERCLOCKWISE from the upper right: I, II, III, IV.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A point starts in Quadrant I at (4, 6). Describe a move that puts it in Quadrant III. What must happen to both coordinates? Can you give the exact new coordinates after your move?

Model answer: Quadrant III is the bottom-left region, so both coordinates have to end up negative. Starting at (4, 6), you need to move more than 4 units left to push x below 0 and more than 6 units down to push y below 0. Moving 6 left and 8 down, for example, lands on (-2, -2), which is Quadrant III because x and y are both negative.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.